

Qualità - 2: Teoria della Misurazione, Ridondanza & Aggregazione

Stefano Mizzaro

DMIF – UNIUD

mizzaro@uniud.it

Social Computing - Lezione 22

19 dicembre 2025

Riassunto

- 18. Intro Crowdsourcing
 - Def., Es.
- 19. Tipologie di Crowdsourcing
 - Varie classificazioni
- 20. (Micro)task design
 - Come progettare un task, processo, metodologia
- 21. Qualità - 1
 - Task design, controlli, ridondanza, aggregazione

Calendario lezioni (?)

«È difficile fare
previsioni, soprattutto
sul futuro» [Bohr ?]

- ✓ 18. Lu 1/12: Intro Crowdsourcing
- ✓ 19. Ve 5/12: Tipologie/classificazione crowdsourcing
- ✓ 20. Ve 12/12: Design
- ✓ 21. Lu 15/12: Qualità 1
 - Task design, Controlli, Ridondanza & Aggregazione
- 22. Ve 19/12 (oggi): Qualità 2
 - Teoria della misurazione
 - Ancora aggregazione
- 23. Lu 22/12: Qualità 3
 - Seminario Michael Soprano su Crowd_Frame
 - Accordo
- 24. Ve 9/1:
 - Di nuovo disinformazione
 - Fine!

Tecniche per migliorare / ottenere buona qualità

- Metodologiche
 - Design del task (Pairwise, scale, presentazione, ...)
- Controlli in itinere
 - Test di qualificazione (Qualification test)
 - Controlli sintattici
 - Test nascosti (Gold questions, honey pots, coerenza)
 - Monitoraggio del comportamento (tempo, azioni, log, ...)
- Ridondanza. Aggregare
- Ex-post
 - Pulizia dati (rimozione outlier, ecc.)
 - Aggregazione sofisticata (co-determination algorithms)
 - Sfruttare l'accordo

Obiettivo della lezione di oggi

- Discutere della qualità (parte 2)
- Ridondanza e aggregazione
- Cenni di teoria della misurazione
 - "Come non aggregare"
 - Come evitare errori

Scaletta

- Design del task
 - Controlli in itinere
 - Ridondanza & Aggregazione
 - Teoria della misurazione
 - Aggregazione reloaded
 - Accordo
 - Def., Varie misure
 - Come sfruttarlo per la qualità
 - Un modello più sofisticato
- Lezione scorsa
- Prossima lezione

Scaletta

- Teoria della misurazione
- Aggregazione reloaded
- (Alcune) funzioni di aggregazione

Scaletta

- Teoria della misurazione
- Aggregazione reloaded
- (Alcune) funzioni di aggregazione



- Measurement
 - assignment + homomorphism
- Scales (N, O, I, R)
- Permissible transformations
- Meaningful statements
- Legit operations & statistics

Measurement

- Definition: A process aimed at determining a relationship between a [physical] quantity and a unit of measurement
 - In realtà anche quantità non fisiche (percezioni, opinioni, ...)
- Typically, one **assigns numbers to objects/events**
 - (che è quello che spesso fanno i worker!)
- Studied in **Measurement Theory**
 - Reasonably settled



WIKIPEDIA
The Free Encyclopedia

Main page
Contents
Featured content
Current events
Random article
Donate to Wikipedia

Interaction
Help
About Wikipedia
Community portal
Recent changes
Contact page

Toolbox
What links here
Related changes
Upload file
Special pages
Permanent link
Page information

Article Talk

Read Edit source View history

Search

Measurement

From Wikipedia, the free encyclopedia

"Mensuration" redirects here. For the musical notation, see [Mensural notation](#).

Measurement (from [Old French](#), *mesurement*) is the assignment of numbers to objects or events.^[1] It is a cornerstone of most [natural sciences](#), [technology](#), [economics](#), and [quantitative research](#) in other social sciences.

Any measurement can be judged by the following meta-measurement criteria values: [level of measurement](#) (which includes [magnitude](#)), dimensions ([units](#)), and [uncertainty](#). They enable comparisons to be done between different measurements and reduce confusion. Even in cases of clear qualitative similarity or difference, increased precision through quantitative measurement is often preferred in order to aid in replication. For example, different colours may be [operationalized](#) based either on [wavelengths](#) of [light](#) or (qualitative) terms such as "green" and "blue" which are often interpreted differently by different people. The science of measurement is called [metrology](#).

Contents [hide]

- Standardization of measurement units
 - Standards
- Units and systems
 - Imperial system
 - Metric system
 - International System of Units
 - Converting prefixes
 - Length
 - Some special names



Defining Measurement - 1

- Assignment
- Given a set of objects $\mathcal{D}$

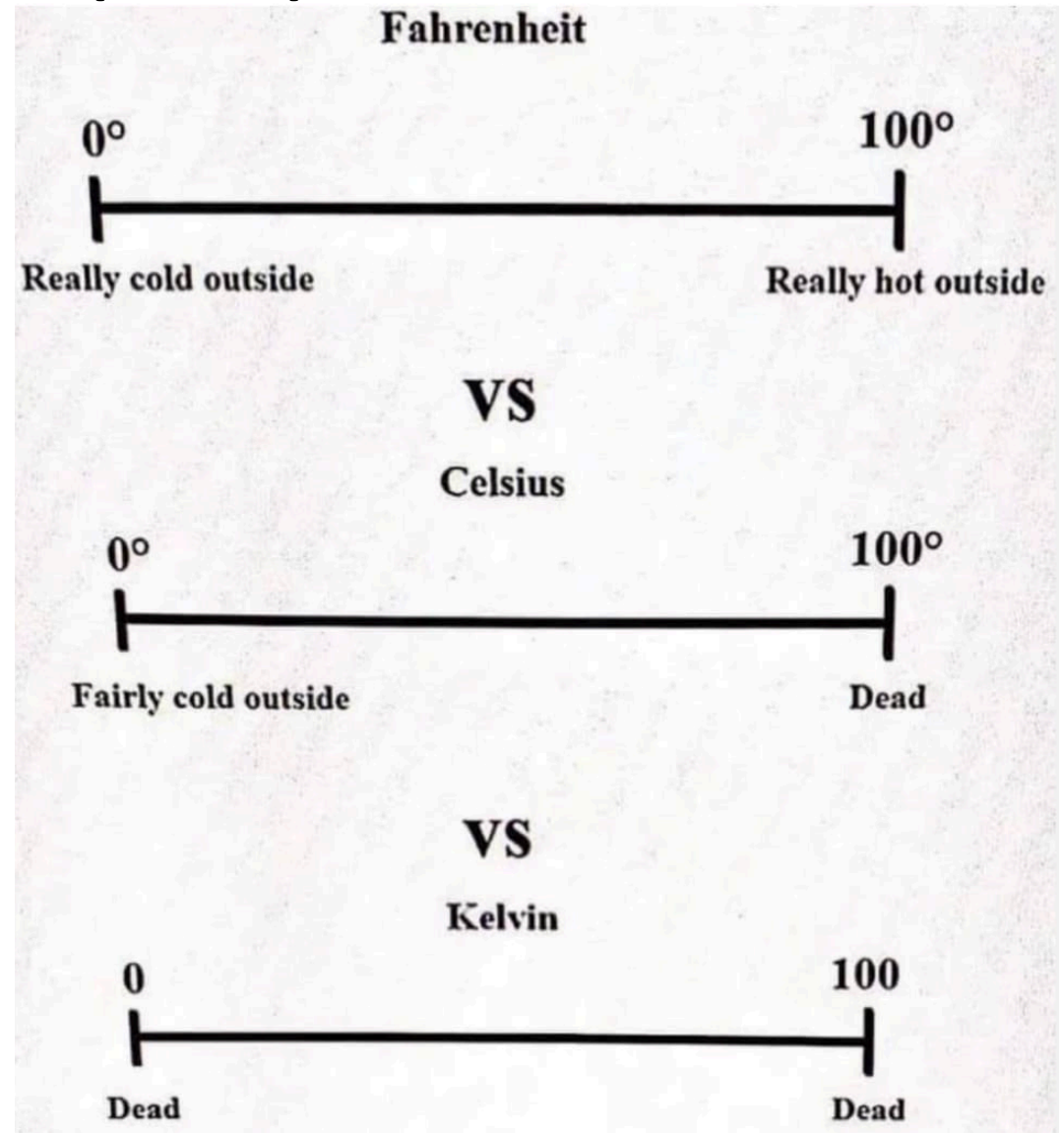
***Definition 1 (Assignment)** An assignment $\omega \in \Omega$ (we denote with Ω the set of all possible assignments) is a function that assigns values to objects in a set $\mathcal{D}$:*

$$\omega : \mathcal{D} \longrightarrow \mathbb{R}.$$

- Tutto qui?!
- Measurement = Assignment of a real number?
- Eh no...

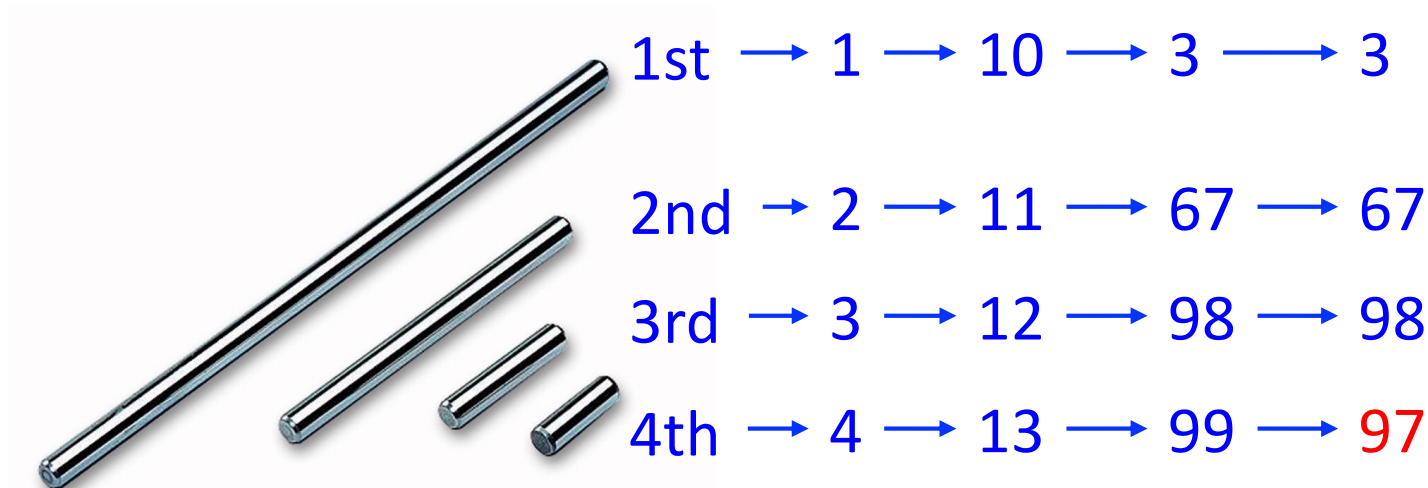
Alcune questioni (1/5)

- Numeri diversi per "misurare la stessa cosa"?!
 - Lunghezza: Metri vs. Centimetri vs. Pollici
 - Temperatura: Celsius (gradi centigradi) vs. Fahrenheit vs. Kelvin
 - Operazioni di conversione, "trasformazione"
- Sono la "stessa misura"?
- "Stessa misura" vs. "Misura equivalente"



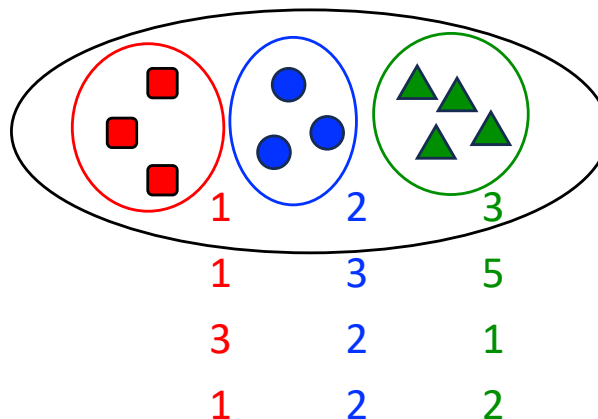
Alcune questioni (2/5)

- Confronto, ordinamento
 - 1o, 2o, 3o, ...: è sempre una misura?
 - E poi cambio i numeri: 1, 2, 3, ... $\rightarrow$ 10, 11, 12, ... o anche 3, 67, 98, ...
- Operazioni di trasformazione ($\neq$ da prima)
- "Stessa misura" vs. "Misura equivalente"



Alcune questioni (3/5)

- Colori (nazionalità, ...)
 - Classifico oggetti in colori (nazionalità, ...): è sempre una misura?
 - Poi do un numero ai colori (blu = 1, rosso = 2, ecc.): è sempre una misura?
 - E se poi cambio i numeri (trasformazione)?! Non è più la stessa, ma è equivalente? Quando?



Alcune questioni (4/5)

- Aggregazioni
 - Temperatura media degli ultimi 10gg: OK
 - Posizione media?
 - Colore medio??

Alcune questioni (5/5)

- "Doppio":
 - OK per lunghezza: A è lungo il doppio di B (m, cm, km, ...)
 - KO per colori: 2 (rosso) è il doppio di 1 (blu)?
 - Ma per temperatura? A è caldo il doppio di B??
- Insomma, ci sono un po' di "cose strane", da capire
- Cerchiamo di essere più sistematici

Defining Measurement - 2

- To be a measurement, an assignment must be a homomorphism

***Definition 2 (Measurement)** Let $\langle \mathcal{D}, R_{\mathcal{D}} \rangle$ be an empirical relational structure and $\langle \mathbb{R}, R_{\mathbb{R}} \rangle$ be a numerical relational structure.² A measurement $\psi \in \Psi$ (we denote with $\Psi \subset \Omega$ the set of all possible measurements) is an assignment of objects in a set $\mathcal{D}$*

$$\psi : \mathcal{D} \longrightarrow \mathbb{R}$$

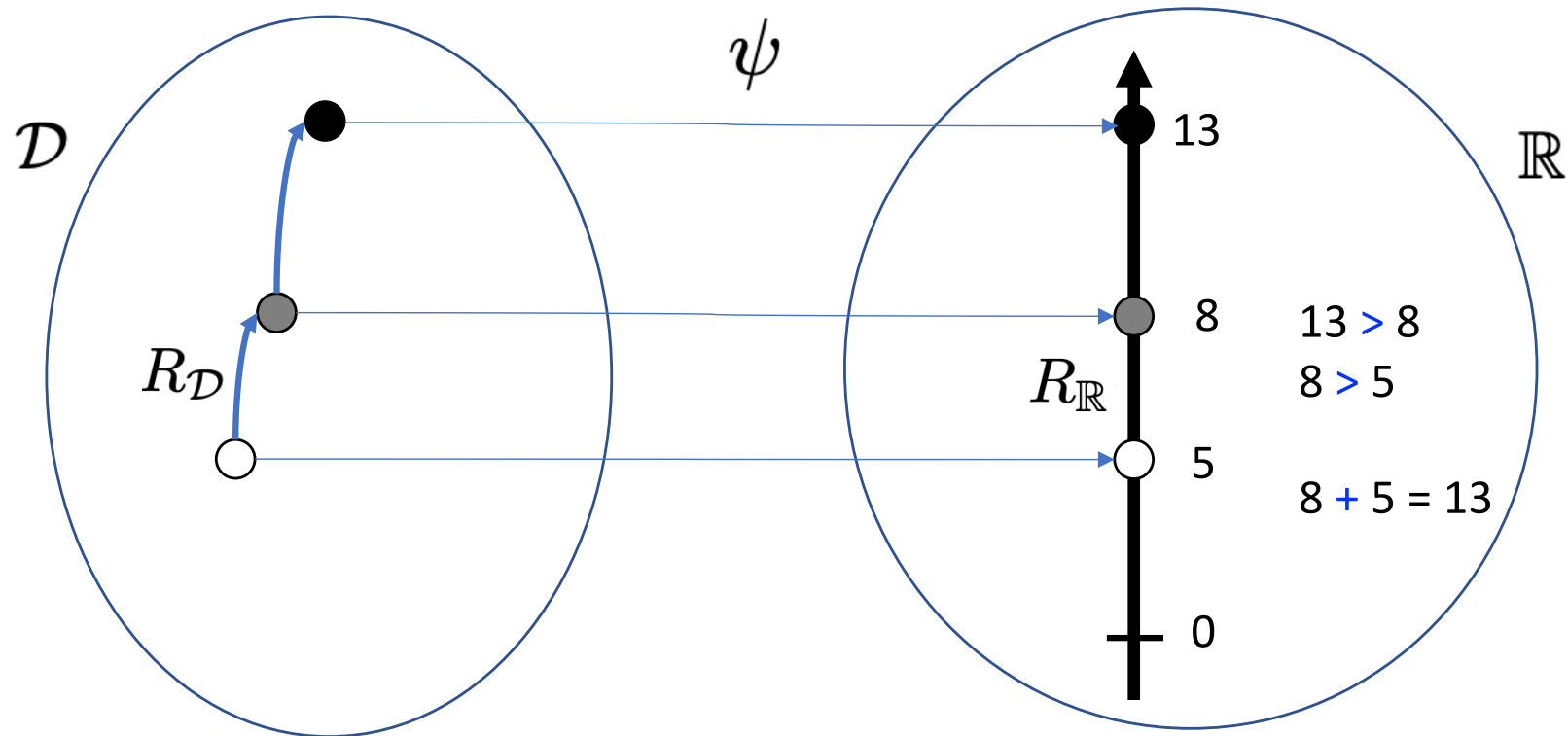
that is a homomorphism between the empirical relational structure and the numerical relational structure, i.e., $\forall x, y \in \mathcal{D} (R_{\mathcal{D}}(x, y) \implies R_{\mathbb{R}}(\psi(x), \psi(y)))$.

Huh? 🤔 🤔

- «I matematici sono come i francesi: se si parla con loro, traducono nella loro lingua, e diventa subito qualcosa di diverso.»

Johann Wolfgang von Goethe

Esempio



Empirical relational structure

Numerical relational structure

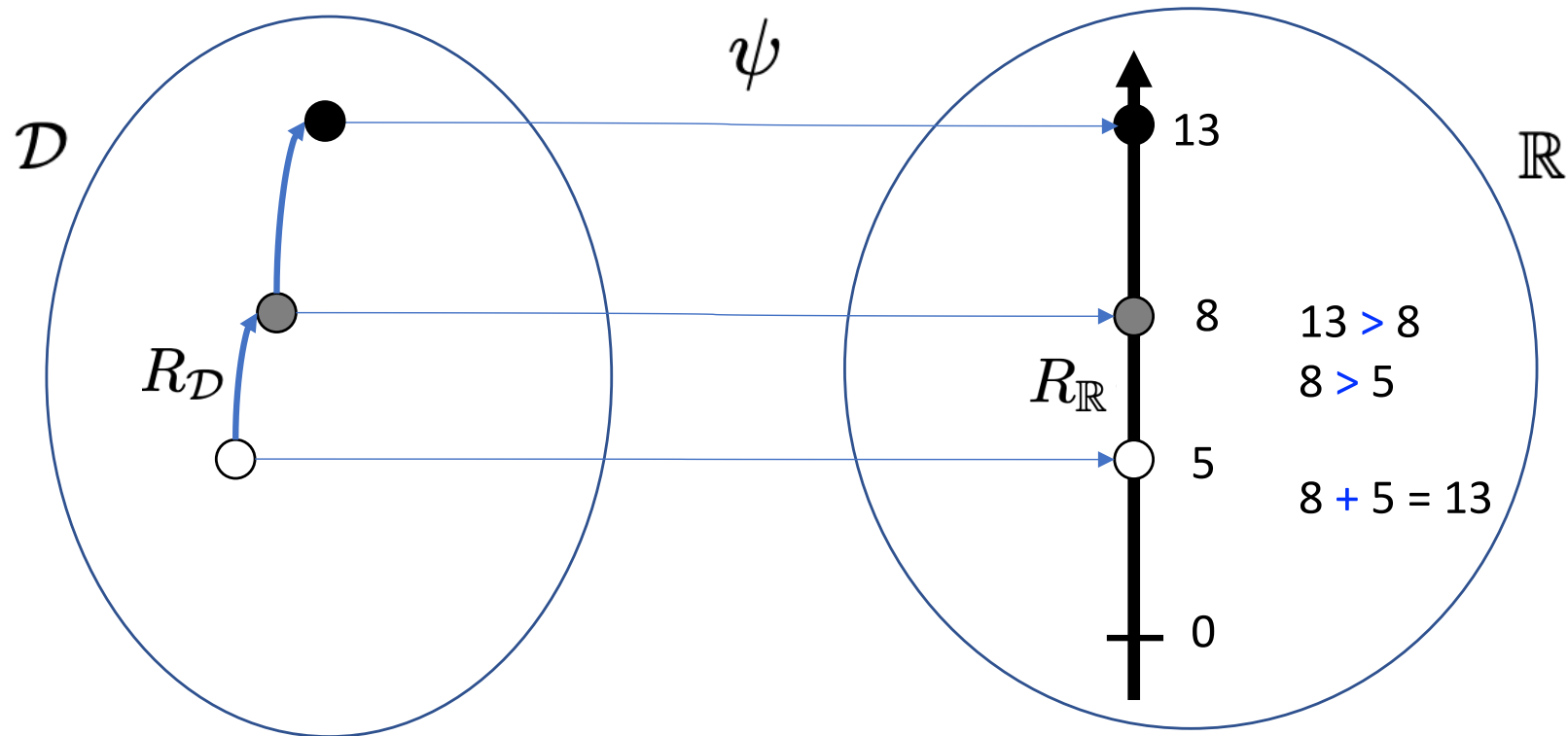
Homomorphism:

- "Heavier than" $\rightarrow$ $>$
- "Agglomeration" $\rightarrow$ $+$

Esempio

- Il nero pesa più del grigio che pesa più del bianco
 - $13 > 8 > 5$
- Il peso del grigio e del bianco insieme è uguale al peso del nero
 - $8 + 5 = 13$
- Qs. proprietà valgono in qualsiasi unità di misura (g, Kg, once, ...)
- Idem per lunghezza:

Esempio



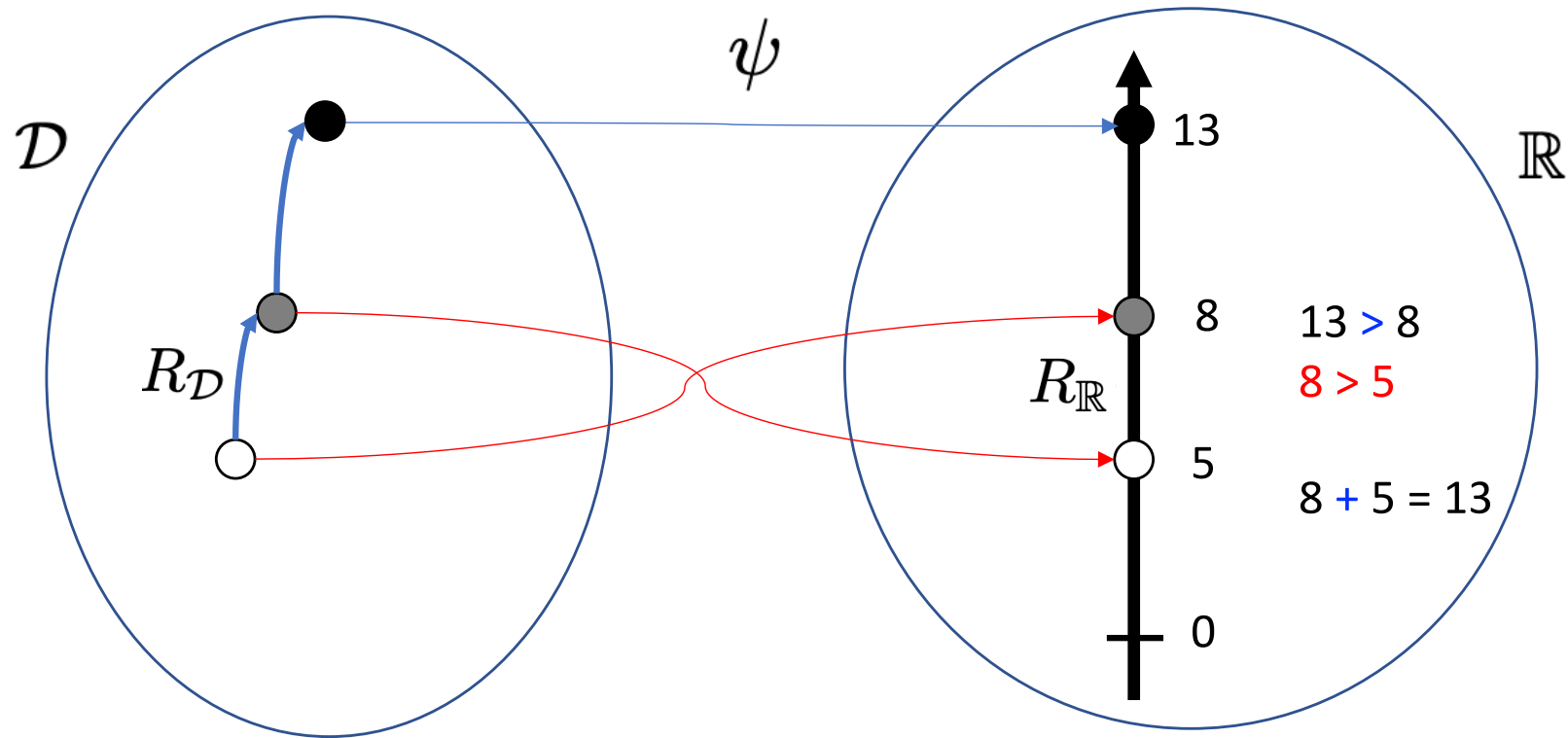
Empirical relational structure

Numerical relational structure

Homomorphism:

- "Longer than" $\rightarrow$ $>$
- "Concatenation" $\rightarrow$ $+$

Esempio



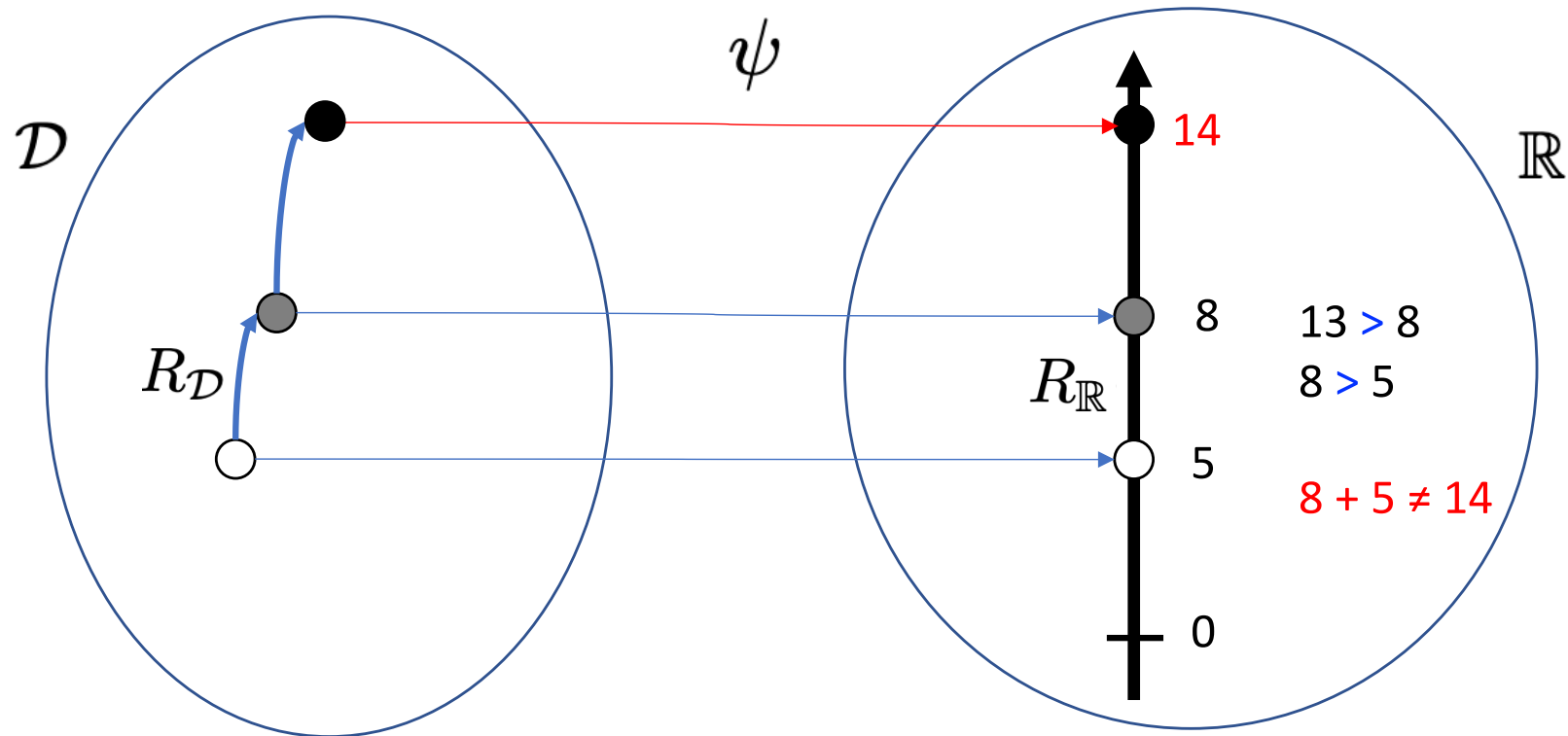
Empirical relational structure

Numerical relational structure

Still an assignment, but no homomorphism:

- "Longer than" $\rightarrow$ $>$
- "Concatenation" $\rightarrow$ $+$

Esempio



Empirical relational structure

Numerical relational structure

Still an assignment, but no homomorphism:

- "Longer than" $\rightarrow$ $>$
- "Concatenation" $\rightarrow$ $+$

Measurement Theory

- One important concept: which numbers can I select when measuring? What properties do they have?
- Which **measurement scale**?
 - (or “**level**”)
- E.g., to measure length:
 - Meters
 - Inches
 - "Longer than" (?)
 - "Heavy" vs. "Light" (??)



SCIENCE

Vol. 103, No. 2684

Friday, June 7, 1946

On the Theory of Scales of Measurement

S. S. Stevens

Director, Psycho-Acoustic Laboratory, Harvard University

FOR SEVEN YEARS A COMMITTEE of the British Association for the Advancement of Science debated the problem of measurement. Appointed in 1932 to represent Section A (Mathematical and Physical Sciences) and Section J (Psychology), the committee was instructed to consider and report upon the possibility of "quantitative estimates of sensory events"—meaning simply: Is it possible to measure human sensation? Deliberation led only to disagreement, mainly about what is meant by the term measurement. An interim report in 1938

by the formal (mathematical) properties of the scales. Furthermore—and this is of great concern to several of the sciences—the statistical manipulations that can legitimately be applied to empirical data depend upon the type of scale against which the data are ordered.

A CLASSIFICATION OF SCALES OF MEASUREMENT

Paraphrasing N. R. Campbell (Final Report, p. 340), we may say that measurement, in the broadest sense, is defined as the assignment of numerals to objects or events according to rules. The fact that

Measurement scales

- Standard set of scales:
 1. Nominal
 2. Ordinal
 3. Interval
 4. Ratio
- (other proposals exist)

Perhaps not so well settled...

QUANTITATIVE METHODS IN PSYCHOLOGY

Measurement Scales and Statistics: A Clash of Paradigms

Joel Michell
University of Sydney
Sydney, New South Wales, Australia

(1986)

The “permissible statistics” controversy stems from a clash of different theories or paradigms of measurement. Three theories are identified: the representational, the operational, and the classical. In each case the relation between measurement scales and statistical procedures is explored. The representational theory implies a relation between measurement scales and statistics, though not the one mentioned by Stevens or his followers. The operational and classical theories, for different reasons, imply no relation between measurement scales and statistics, contradicting Stevens’s prescriptions. A resolution of this issue depends on a critical evaluation of these different theories.

And indeed

- Nicholas Chrisman
- 1998
- Proposes 10 scales, not just 4

ABSTRACT: Stevens' measurement levels (nominal, ordinal, interval and ratio) have become a familiar part of cartography and GIS. These levels have been accepted unquestioned from publications in social sciences dating from the 1940s and 1950s. The Stevens taxonomy has been used to prescribe appropriate symbolism (or analytical treatment) to each scale of measurement. This paper reviews the process by which these levels became a part of cartography, as well as subsequent literature that cartographers have all but ignored over the intervening four decades. The paper concludes that the four levels of measurement are not adequate to cover the circumstances of cartography, and that attribute issues alone do not provide a sufficient guide to symbolism or analytical treatment. A broader framework for measurement must be considered, including the relationships of control that constrain variation in one component to permit measurement of another. An informed use of tools does not depend on numbers alone, but on the whole "measurement framework," the system of objects, relationships and axioms implied by a given system of representation.

How Measurement Levels Reached Cartography

The approach to measurement in certain social sciences (including geography) is still strongly influenced by Stevens' (1946) paper *Science*. Stevens' levels of measurement form an important foundation for textbooks in spatial analysis (Unwin 1981) and in cartography (Robinson et al. 1985; Muehrcke and Muehrcke 1978; Dent 1993). While there has been continued development of the theory of measurement (Khurshid and Sahai 1993), debate on measurement in cartography and GIS literature remains stuck in 1946. This phenomenon is interesting because it is followed by an absorption of a new set of ideas, and resistance to further changes. It also is important because the measurement concepts serve as foundations for instruction. This first section will review the process leading up to Stevens' paper and how it diffused into cartographic practice.

History of Theories of Measurement

The "classical" school of measurement has its origins in Aristotle's metaphysics and continues through Newton to include the physical sciences

at the end of the nineteenth century. In the classical view, measurement discovered the numerical ratio between a standard object and the measured.

Clifford, a mathematician in the nineteenth century, used a definition of measurement not very different from Euclid's: "every quantity is measured by the ratio which it bears to some fixed quantity called the unit" (Clifford 1870). The property seen to be "inherent" in the object, and the relationship took on all the mathematical properties of numbers without question (Michell 1993).

Let us take the attribute "length." Every entity in space can be measured by comparing its length to some other length. If we adopt a "standard" measuring rod, the ratio between the length of the rod and the objects measured embodies a number implicating the relationship. The realm of numbers is then, in turn, seen to be derived from these basic relationships of ratios.

Through a series of theoretical developments, this classical approach was overthrown and completely replaced by a "representational" school in the first half of the twentieth century (Michell 1993). Representationalists impose a firm separation between the empirical world and numbers, as expressed by Bertrand Russell (Russell 1897; 1903). In the place of ratios and rational numbers as the core of mathematics, set theory permitted a more generic range of relationships. According to Russell:

Measurement of magnitudes is, in its most general sense, any method by which a unique and reciprocal correspondence is established between all or some of the

Nicholas Chrisman is Associate Professor in the Department of Geography, Room 408, Smith Hall, University of Washington, Box 353550, Seattle, Washington 98195, U.S.A. E-mail: <chrisman@u.washington.edu>.

Anyway "Good Old Fashioned" Measurement scales

1. Nominal
2. Ordinal
3. Interval
4. Ratio



4. Ratio scale

- Height
 - "I'm twice taller than him"
- Money
 - "He is twice richer than me"
- (Both **how much** & **how many**, continuous & discrete)
- Age
 - "I'm twice older than you"
 - Years, months (* 12), days (* 365), ...
- Zero
 - Height, Richness, Age start from zero!
- But not all measures are like that:
 - "Today is twice as hot as yesterday"?

Trasformazioni permissibili:
misura * costante,
 $y = f(x)$, dove $f(x) = a * x$, $a > 0$

GUYS, IF $0^{\circ}\text{C} = 32^{\circ}\text{F}$
THEN $0^{\circ}\text{C} + 0^{\circ}\text{C} = 64^{\circ}\text{F}$

u/karsbil10



3. Interval scale

- "Today is twice as hot as yesterday"?
- In the last two days we had the same increase of 5°C in temperature
 - The difference between today and yesterday temperature is the same as ...
- Dates are another example
 - Same difference: OK ($2019 - 2017 = 2006 - 2004 = 2$ years)
 - Ratio: KO (year 2000 is not $2 * 1000$)
 - (ratios of differences: OK
 - $2021 - 2017 = 4$ years; $2006 - 2004 = 2$ years; 4 is the double of 2)

Celsius $\rightarrow$ Fahrenheit

$$\begin{aligned}^{\circ}\text{F} &= (^{\circ}\text{C} \times 1.8) + 32 = \\ &= (^{\circ}\text{C} \times \frac{9}{5}) + 32\end{aligned}$$

Trasformazioni permissibili:
 $y = f(x)$, dove $f(x) = a x + b$, $a > 0$

2. Ordinal scale

- A measure is not an amount but a **rank**
 - The numbers are rank positions (1st, 2nd, ...)
- It is a form of measurement!
- Ex:
 - Today = 1, Yesterday = 2
 - "Today is hotter than yesterday"
 - Lines ranked according to their length
- It does **not** mean that:
 - the first is twice as long than the second
 - the length difference between the 1st and the 2nd is the same as the 2nd and the 3rd



Trasformazioni permissibili:
 $y = f(x)$,
dove f monotona crescente

1. Nominal scale

- Qualitative
- Categories
- Names, gender, nationality, colors, ...
- The numbers are class identifiers
- Ex.:
 - 0 = Cold, 1 = Hot
 - 0 = Male, 1 = Female
 - 1 = Red, 2 = Blue, 3 = Green, 4 = White
- Can be Dichotomous or Non-dichotomous
- No assumptions on ratios, distances, ranks.

Trasformazioni permissibili:
 $y = f(x)$,
dove f iniettiva

Permissible transformations

- **Given a scale, the measurement can be transformed to obtain an equivalent measurement**

		(increasing)		(a > 0)
	1-to-1	Monotonic	$a \cdot x + b$	$a \cdot x$
Nominal	✓	✓	✓	✓
Ordinal	✗	✓	✓	✓
Interval	✗	✗	✓	✓
Ratio	✗	✗	✗	✓

Meaning

- If you **transform** the measure, you are still measuring the same thing (in the "empirical relational structure")
 - Nationality
 - Rank
 - Temperature
 - Money

Meaningful statements

- Some statements on a measurement make sense, others do not

Examples

- Age
 - years * 12 = months
 - 40 years = 2 * 20 years
 - My age is the double is yours
- Nominal scale for nationality
 - Greek = 1
 - Italian = 2
 - Spanish = 3
 - Japanese = 4
 - ...
 - Italian is the double of Greek? Uh?!?
 - $3 - 1 = 4 - 2$? Uh?!?
- Whereas interval scale for temperature °C
 - $30^{\circ} - 10^{\circ} = 40^{\circ} - 20^{\circ}$: ok

Meaningful statements

- Of course, the wrong statements are wrong!
 - Ex.: $30^\circ - 10^\circ = 20^\circ - 5^\circ$ (is wrong!)
- **Given a scale**, only some (of the correct) statements make sense (are **meaningful**)
 - $=, \neq, <, >$
 - Ex.:
 - Numeric codes for colors (1 = Red, 2 = Blue, 3 = Green, 4 = White)
 - $1 = 1, 1 \neq 2$... OK
 - $1 < 2, 2 - 1 = 3 - 2, 4 / 2 = 2 / 1$: KO
- Their truthfulness (falsehood) holds even after permissible transformations

Legit statements / operations

- **Given a scale**, only some operations make sense (are **legit**) (are "preserved" by permissible transformations)
 - Relational: $=$, $\neq$, $>$, $<$
 - Arithmetic: $+$, $-$, $*$, $/$
 - Statistic: Mean, median, mode, ...
- Ex.:
 - Average height, weight, ...: OK
 - Average gender: KO

Legit relational/arithmetic operations

	$=, \neq$	$>, <$	$+, -$	$\times, \div$
Nominal	✓	✗	✗	✗
Ordinal	✓	✓	✗	✗
Interval	✓	✓	✓	✗
Ratio	✓	✓	✓	✓

Example

- I measure the length of two items on some ratio scale
 - $a = \text{length}(\text{item 1}) = 4$
 - $b = \text{length}(\text{item 2}) = 8$
 - It holds that $8 = 2 * 4$
 - $b = 2 * a$, I'm using legit operations ($=$, $*$)
- If I apply $f(x) = 10 * x$ (permissible transformation)
 - $4 \rightarrow 40, 8 \rightarrow 80$
 - $80 = 2 * 40$, it still holds that $b = 2 * a$
- If I apply $f(x) = 10 * x + 1$ (non permissible transformation)
 - $4 \rightarrow 41, 8 \rightarrow 81$
 - $81 \neq 2 * 41$, this breaks the relation $b = 2 * a$

Do not confuse

- Permissible transformations
 - transformations that I can apply to a measurement and still get "the same" (an equivalent) measurement
- Legit operations
 - things that I can use to define relations between measurements
- From the latter one can derive →

Legit statistics

	Mode	Median	Mean	
			Arithmetic	Geometric, Harmonic
Nominal	✓	✗	✗	✗
Ordinal	✓	✓	✗	✗
Interval	✓	✓	✓	✗
Ratio	✓	✓	✓	✓

Legit statistics to be used according to measurement theory

	Mode	Median	Mean	
			Arithmetic	Geometric, Harmonic
Nominal	✓	X	X	X
Ordinal	✓	✓	X	X
Interval	✓	✓	✓	X
Ratio	✓	✓	✓	✓

Use these!

But...

	Mode	Median	Mean	
			Arithmetic	Geometric, Harmonic
Nominal	✓	✗	✗	✗
Ordinal	✓	✓	✗	✗
Interval	✓	✓	✓	✗
Ratio	✓	✓	✓	✓

Legit, but info loss!

Examples

- Nationality, mode: ok
- Mean rank? Uh?!
 - (think of ranking ten students...)
- Mean temperature: ok

Ex.: Semantic differential

Table 2.6. Typical Semantic Differential*

	Agree Strongly	Agree Slightly	Agree with Neither	Agree Slightly	Agree Strongly	
Financially, we, as a family, are better off than we were a year ago	☐	☐	☐	☐	☐	Financially, we, as a family, are less well off than we were a year ago
	1,	2,	3,	4,	5,	6, 7?

- Average??
- Agh, no...
- Now, that is a canonical / "extremist" / integralist position

Likert scales, levels of measurement and the “laws” of statistics

Geoff Norman

Received: 22 January 2010 / Accepted: 22 January 2010
© Springer Science+Business Media B.V. 2010

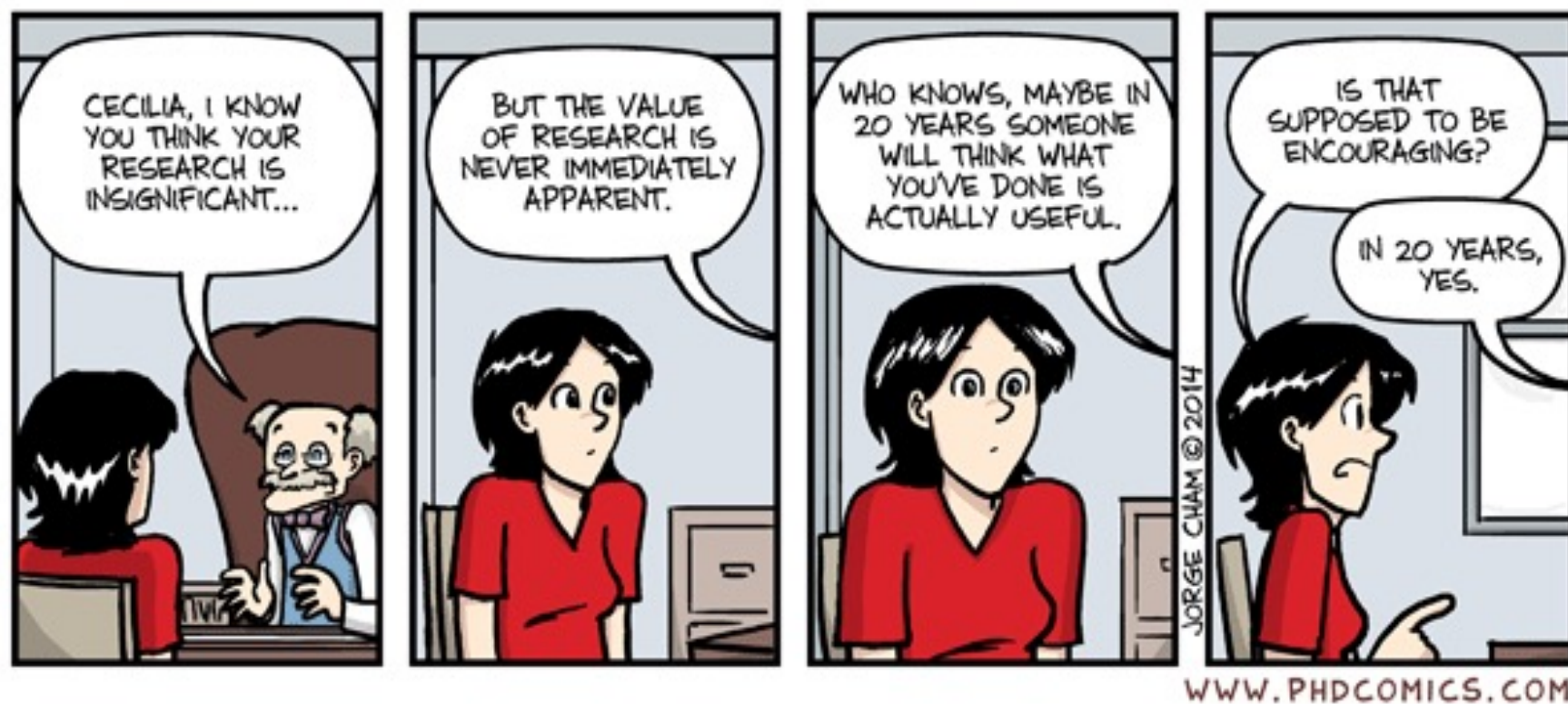
Abstract Reviewers of research reports frequently criticize the choice of statistical methods. While some of these criticisms are well-founded, frequently the use of various parametric methods such as analysis of variance, regression, correlation are faulted because: (a) the sample size is too small, (b) the data may not be normally distributed, or (c) The data are from Likert scales, which are ordinal, so parametric statistics cannot be used. In this paper, I dissect these arguments, and show that many studies, dating back to the 1930s consistently show that parametric statistics are robust with respect to violations of these assumptions. Hence, challenges like those above are unfounded, and parametric methods can be utilized without concern for “getting the wrong answer”.

Summary

- Measurement (assignment + homomorphism)
- Scales (N, O, I, R)
- Permissible transformations
- Meaningful statements
- Legit operations & statistics

Why are you telling me this?





O magari per avere qualche domanda da farvi all'esame 🤔😬😓

Scaletta

- Teoria della misurazione
- **Aggregazione reloaded**
- (Alcune) funzioni di aggregazione

A cosa ci serve?

- A ragionare sull'aggregazione
- Riprendiamo un esempio dalla lezione scorsa
- Relevance assessment binario

Document Relevance Evaluation

Instructions

Evaluate the relevance of a document to the space program topic. A document is relevant if it would be useful if you were writing a report on the space program topic. Each document should be judged on its own merits. That is, a document is still relevant even if it is the thirtieth document you have seen with the same information.

Tips

- Payment based on quality of the work completed. Please follow the instructions.
- Be consistent in your judgments.
- A document should not be judged as relevant or irrelevant based only on the title of the document. You must read the document

Task

Please evaluate the relevance of the following document about the space program topic. To be relevant, a document must discuss the goals or plans of a space program or space project and identify the organization(s) sponsoring the program. The space program is a coherent set of initiatives to exploit outer space. The space project is a specific mission to exploit outer space.

\$DOCUMENT\$

Please rate the above document according to its relevance to the space program topic as follows. Note that the task is about how relevant to the topic the document is.

- () Relevant. A relevant document for the topic.
- () Not relevant. The document is not good because it doesn't contain any relevant information.

Esempio

- v_{ij} è
 - o Relevant
 - o Not relevant
- Idem per cv_j

	t_1	t_2	t_3	...	t_n
w_1	v_{11}	v_{12}	v_{13}		v_{1n}
w_2	v_{21}		v_{23}	...	v_{2n}
w_3	v_{31}	v_{32}		...	v_{3n}
...				...	
w_m		v_{m2}	v_{m3}		

Aggr	av_1	av_2	av_3	...	av_n
------	--------	--------	--------	-----	--------

Truth	cv_1	cv_2	cv_3	...	cv_n
-------	--------	--------	--------	-----	--------

Esempio

- v_{ij} è
 - o Relevant (R)
 - o Not relevant (N)
- Idem per cv_j

	t_1	t_2	t_3	...	t_n
w_1	R	N	R		R
w_2	R		N	...	N
w_3	R	N		...	N
...				...	
w_m		N	R		

Aggr	av_1	av_2	av_3	...	av_n
------	--------	--------	--------	-----	--------

Truth	R	N	R	...	N
-------	---	---	---	-----	---

Esempio

- v_{ij} è
 - o Relevant (1)
 - o Not relevant (0)
- Idem per cv_j
- E av_j ?

	t_1	t_2	t_3	...	t_n
w_1	1	0	1		1
w_2	1		0	...	0
w_3	1	0		...	0
...				...	
w_m		0	1		

Aggr	av_1	av_2	av_3	...	av_n
------	--------	--------	--------	-----	--------

Truth	1	0	1	...	0
-------	---	---	---	-----	---

Esempio

- v_{ij} è
 - o Relevant (1)
 - o Not relevant (0)
- Idem per cv_j
- E av_j ?
 - Dipende...
 - Moda (MV)?

	t_1	t_2	t_3	...	t_n
w_1	1	0	1		1
w_2	1		0	...	0
w_3	1	0		...	0
...				...	
w_m		0	1		

Aggr	1	0	1	...	0
------	---	---	---	-----	---

Truth	1	0	1	...	0
-------	---	---	---	-----	---

Esempio

- v_{ij} è
 - o Relevant (1)
 - o Not relevant (0)
- Idem per cv_j
- E av_j ?
 - Dipende...
 - Moda (MV)?
 - Media?

	t_1	t_2	t_3	...	t_n
w_1	1	0	1		1
w_2	1		0	...	0
w_3	1	0		...	0
...				...	
w_m		0	1		

Aggr	1	0	2/3	...	1/3
------	---	---	-----	-----	-----

Truth	1	0	1	...	0
-------	---	---	---	-----	---

Scala nominale

- 2 valori (categorie): N e R
 - Oppure 0 e 1
- ...
- Qui è facile e cambia poco...
 - Moda
 - Media non legittima ma se arrotondata = Moda
- Ma in altri casi la scelta è più delicata
 - Discutiamo l'es. delle **Categorie ordinali**
 - che capita spesso
 - ... e che abbiamo già incontrato... (3 esempi)

Document Relevance Evaluation

Instructions

Evaluate the relevance of a document to the given topic. A document is relevant if it would be useful if you were writing a report on such topic. Each document should be judged on its own merits. That is, a document is still relevant even if it is the thirtieth document you have seen with the same information.

Tips

- Payment is based on quality of the work completed. Please follow the instructions.
- Be consistent in your judgments.
- A document should not be judged as relevant or irrelevant based only on the title of the document. You must read the document.

Task

Please evaluate the relevance of the following document about \$TOPIC\$.

\$DOCUMENT\$

Please rate the above document according to its relevance to \$TOPIC\$ as follows. Note that the task is about how relevant to the topic the document is.

- () **Excellent.** The document discusses the themes of the topic exhaustively.
- () **Good.** The document contains information but the presentation is not exhaustive or there might be a better document.
- () **Fair.** The document only points to the topic. It does not contain more information than the topic description or portions of the text are inaccurate.
- () **Not relevant.** The document is not good because it does not contain any relevant information.

Please justify your answer. We appreciate your comments.

[InputBox]

Search Engine Relevance Evaluation

Please look at this web page and answer how relevant it is to the query.

Task

How relevant is the following webpage to the query
\$QUERY\$.

\$WEBPAGE\$

- () **Perfect**. Exactly and completely what the user wanted, there could be no better result.
- () **Excellent**. Exactly what the user wanted, but there may be other results that as good or better
- () **Good**. Mostly what the user wanted, but not exactly.
- () **Fair**. Loosely related to what the user wanted.
- () **Bad**. Certainly not what the user wanted

Please let us know if you have any comments.

[InputBox]

6 valori

- Pants On Fire
- False
- Barely True
- Half True
- Mostly True
- True

Statement 2 of 8

Statement

If I were not a sumo wrestler, I would be a football player.

Speaker: Abraham Lincoln
Speaker's Job: Former U.S. President
Context: Public appearance

Please input the truth level for the above statement as one from the six labels provided

- ☐ Pants On Fire
- ☐ False
- ☐ Barely True
- ☐ Half True
- ☐ Mostly True
- ☐ True

Please provide a valid source (URL) to motivate your choice:

Example: https://en.wikipedia.org/wiki/We_choose_to_go_to_the_Moon

Please provide a justification to motivate your choice:

Justification

Back Next

Proviamo ad aggregare

- 6 categorie
 - Pants On Fire
 - False
 - Barely True
 - Half True
 - Mostly True
 - True
- Prima erano 2
 - Relevant
 - Not relevant
- Prima usavamo la Moda, usiamola anche adesso

Proviamo ad aggregare (Moda)

- Pants On Fire → P
- False → F
- Barely True → B
- Half True → H
- Mostly True → M
- True → T

	t_1	t_2	t_3	...	t_n
w_1	P	P	P		F
w_2	P		P	...	F
w_3	P	P		...	F
...				...	
w_m		F	T		

Aggr	P	P	P	...	F
------	---	---	---	-----	---

Truth	P	P	P	...	P
-------	---	---	---	-----	---

Cosa non ci piace?

- Potenziale perdita di info, come prima
 - colonna con 51 T e 49 M = 99 T e 1 M
- Ma soprattutto
 - Errori piccoli = errori grandi
 - F e T in t_2 e t_3
- t_2 e t_3 sono = ma sono $\neq$
 - E se 51 P e 49 T in t_3 ?!
 - Sempre P , ma ben $\neq$ da 99 P e 1 T ... o da 51 P e 49 F o...
- t_3 e t_n difficili da confrontare
- Si perdono informazioni!

	t_1	t_2	t_3	...	t_n
w_1	P	P	P		F
w_2	P		P	...	F
w_3	P	P		...	F
...				...	
w_m		F	T		

Aggr	P	P	P	...	F
------	---	---	---	-----	---

Truth	P	P	P	...	P
-------	---	---	---	-----	---

Proviamo con la media

- Prima erano 2
 - Relevant (1)
 - Not relevant (0)
 - Media **non legittima** ma se arrotondata = Moda
- Anche qui non legittima.
 - Scala ordinale → mediana 🧑🎓🧐🤔
- Ma proviamoci
- Domanda:
 - Che valori numerici usiamo qui?

Che valori nelle celle?

- Pants On Fire → ?
- False → ?
- Barely True → ?
- Half True → ?
- Mostly True → ?
- True → ?

	t_1	t_2	t_3	...	t_n
w_1	v_{11}	v_{12}	v_{13}		v_{1n}
w_2	v_{21}		v_{23}	...	v_{2n}
w_3	v_{31}	v_{32}		...	v_{3n}
...				...	
w_m		v_{m2}	v_{m3}		

Truth	cv_1	cv_2	cv_3	...	cv_n
-------	--------	--------	--------	-----	--------

Che valori nelle celle?

- Pants On Fire $\rightarrow$ 0
- False $\rightarrow$ 1
- Barely True $\rightarrow$ 2
- Half True $\rightarrow$ 3
- Mostly True $\rightarrow$ 4
- True $\rightarrow$ 5

	t_1	t_2	t_3	...	t_n
w_1	v_{11}	v_{12}	v_{13}		v_{1n}
w_2	v_{21}		v_{23}	...	v_{2n}
w_3	v_{31}	v_{32}		...	v_{3n}
...				...	
w_m		v_{m2}	v_{m3}		

Truth	cv_1	cv_2	cv_3	...	cv_n
-------	--------	--------	--------	-----	--------

Che valori nelle celle?

- Pants On Fire $\rightarrow$ 1
- False $\rightarrow$ 0
- Barely True $\rightarrow$ 5
- Half True $\rightarrow$ 4
- Mostly True $\rightarrow$ 3
- True $\rightarrow$ 2

	t_1	t_2	t_3	...	t_n
w_1	v_{11}	v_{12}	v_{13}		v_{1n}
w_2	v_{21}		v_{23}	...	v_{2n}
w_3	v_{31}	v_{32}		...	v_{3n}
...				...	
w_m		v_{m2}	v_{m3}		

Truth	cv_1	cv_2	cv_3	...	cv_n
-------	--------	--------	--------	-----	--------

Che valori nelle celle?

- Pants On Fire → 1
- False → 0
- Barely True → 5
- Half True → 4
- Mostly True → 3
- True → 2

(Beh questi sicuramente no!)

	t_1	t_2	t_3	...	t_n
w_1	v_{11}	v_{12}	v_{13}		v_{1n}
w_2	v_{21}		v_{23}	...	v_{2n}
w_3	v_{31}	v_{32}		...	v_{3n}
...				...	
w_m		v_{m2}	v_{m3}		

Truth	cv_1	cv_2	cv_3	...	cv_n
-------	--------	--------	--------	-----	--------

Che valori nelle celle?

- Pants On Fire → 0
- False → 10
- Barely True → 100
- Half True → 1 000
- Mostly True → 10 000
- True → 100 000

(Questi anche sì!)

	t_1	t_2	t_3	...	t_n
w_1	v_{11}	v_{12}	v_{13}		v_{1n}
w_2	v_{21}		v_{23}	...	v_{2n}
w_3	v_{31}	v_{32}		...	v_{3n}
...				...	
w_m		v_{m2}	v_{m3}		

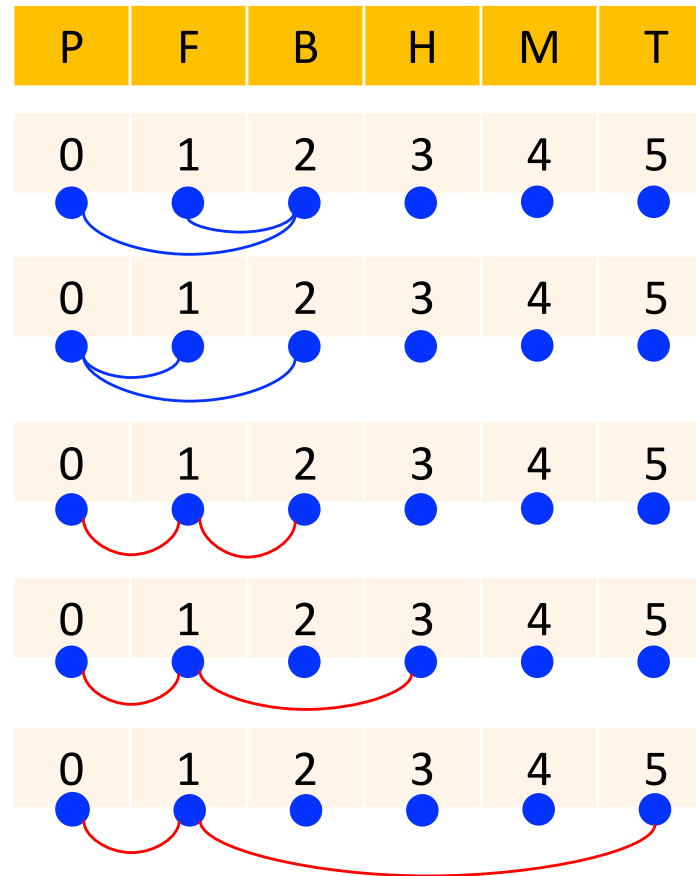
Truth	cv_1	cv_2	cv_3	...	cv_n
-------	--------	--------	--------	-----	--------

Quindi?

- Non è una scala nominale, né a intervalli. È ordinale
 - L'unica cosa certa è l'ordine 🧑🎓🧐🤔
 - Le distanze fra i livelli no! (altrimenti sarebbe una scala a intervalli) 🙋
- Possiamo ipotizzare ("assumere") che 012345 sia ok
- Ma è appunto un'ipotesi di lavoro
- Ci sono cose che possiamo dire con certezza
 - (ordine: qualsiasi trasformazione permissibile non lo cambia)
- e altre no
 - (intervalli: esistono trasformazioni permissibili che lo cambiano)

Cosa possiamo dire con certezza e cosa no

- $2 - 1 < 2 - 0$? OK!
- $1 - 0 < 2 - 0$? OK!
- $1 - 0 = 2 - 1$? Boh?
- $1 - 0 < 3 - 1$? Boh?
- $1 - 0 < 5 - 1$? Boh?



Per convincervi: alternative, incoerenze

- Pants On Fire $\rightarrow 0$
- False $\rightarrow 1$
- Barely True $\rightarrow 2$
- Half True $\rightarrow 3$
- Mostly True $\rightarrow 4$
- True $\rightarrow 5$

- False $\rightarrow 0$
- Barely True $\rightarrow 1$
- Half True $\rightarrow 2$
- Mostly True $\rightarrow 3$
- True $\rightarrow 4$

- False $\rightarrow 0$
- Barely True $\rightarrow 1$
- Half True $\rightarrow 2$
- Mostly True $\rightarrow 3$
- True $\rightarrow 4$
- Definitely True $\rightarrow 5$

- Completely False $\rightarrow 0$
- False $\rightarrow 1$
- Mostly False $\rightarrow 2$
- Mostly True $\rightarrow 3$
- True $\rightarrow 4$
- Completely True $\rightarrow 5$

- Completely False $\rightarrow 0$
- False $\rightarrow 1$
- Mostly False $\rightarrow 2$
- Neither true nor false $\rightarrow 3$
- True $\rightarrow 4$
- Completely True $\rightarrow 5$

(notare gli effetti
delle etichette)

Uhm

- E S100? 🤔

Statement 1 of 8

Statement

Republican (small business tax cut) bill spends \$80 million for better Google searches, not job creation

Speaker: Cory Mason

Speaker's Job: State Assembly member, 62nd District

Context: a news release

Please input the truth level for the above statement as a number in the [0,100] range

Selected Value: 50

Please provide a valid source (URL) to motivate your choice:

Example: https://en.wikipedia.org/wiki/We_choose_to_go_to_the_Moon

Please provide a justification to motivate your choice:

Justification

Next

Beh

- È sempre una scala a categorie ordinali 🧑🎓🧐🤔
- Ma forse con 100 (101) valori, e etichette numeriche, possiamo pensare che sia una scala a intervalli ($12 - 2 = 92 - 82$), "o quasi"
- O forse anche a rapporti ($10 / 2 = 50 / 10$), "o quasi"
- Ma dov'è il limite? 7 valori? 12? 42?
- Buon senso
- Importanza delle etichette
 - (0, 1, 2, 3, 4, 5 ≠
Pants On Fire, False, Barely True, Half True, Mostly True, True)
- Insomma, a essere rigidi...
- ... ma si può essere piu soft, purché **consapevolmente!**

Likert scales, levels of measurement and the “laws” of statistics

Geoff Norman

Received: 22 January 2010 / Accepted: 22 January 2010
© Springer Science+Business Media B.V. 2010

Abstract Reviewers of research reports frequently criticize the choice of statistical methods. While some of these criticisms are well-founded, frequently the use of various parametric methods such as analysis of variance, regression, correlation are faulted because: (a) the sample size is too small, (b) the data may not be normally distributed, or (c) The data are from Likert scales, which are ordinal, so parametric statistics cannot be used. In this paper, I dissect these arguments, and show that many studies, dating back to the 1930s consistently show that parametric statistics are robust with respect to violations of these assumptions. Hence, challenges like those above are unfounded, and parametric methods can be utilized without concern for “getting the wrong answer”.

Scaletta

- Teoria della misurazione
- Aggregazione reloaded
- (Alcune) funzioni di aggregazione
 - Media aritmetica e Media geometrica
 - Outlier e Mediana
 - Media armonica

Ancora sull'Aggregazione

- Sono possibili varie scelte / funzioni
- Media aritmetica (AM)
- Media geometrica (GM)
- Mediana
 - Meno sensibile agli outlier dell'AM
- Media armonica (HM)

	t_1	t_2	t_3	...	t_n
w_1	v_{11}	v_{12}	v_{13}		v_{1n}
w_2	v_{21}		v_{23}	...	v_{2n}
w_3	v_{31}	v_{32}		...	v_{3n}
...				...	
w_m		v_{m2}	v_{m3}		
Aggr	av_1	av_2	av_3	...	av_n

$$av_j = F_{aggr}(\{v_{ij} | w_i \in W_j\}) = F_{aggr}(\{v_{ij} | v_{ij} \neq \perp\})$$

Funzioni di aggregazione

$$av_j = F_{aggr}(\{v_{ij} | w_i \in W_j\}) = F_{aggr}(\{v_{ij} | v_{ij} \neq \perp\})$$

- Moda, A maggioranza, MV (Majority Voting)
- Mediana
- Media aritmetica
- Media geometrica
- Media armonica
- Medie pesate
- ...
- ... non sempre è semplice scegliere

Medie aritmetica e geometrica

Media Aritmetica

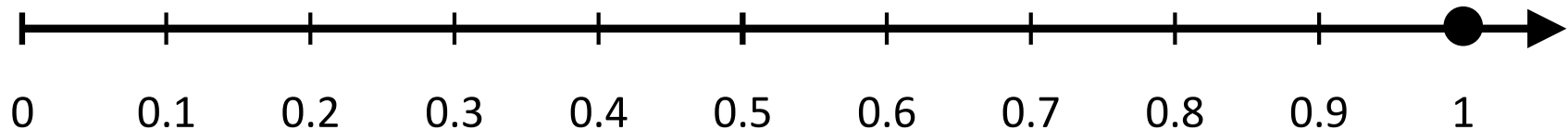
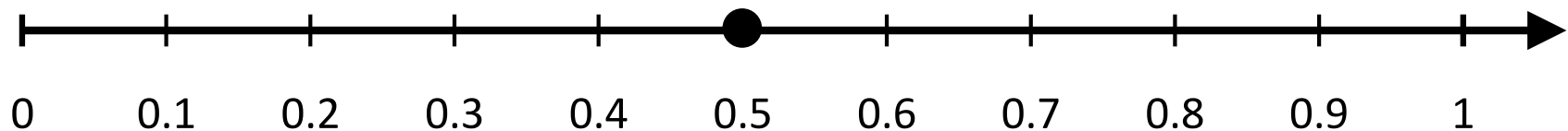
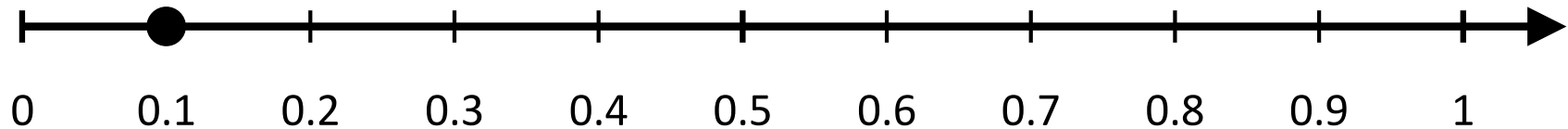
- $\mathbf{x} = \langle x_1, x_2, \dots, x_n \rangle$
- $AM(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n x_i$

Media Geometrica

- $\mathbf{x} = \langle x_1, x_2, \dots, x_n \rangle$
- $GM(\mathbf{x}) = \sqrt[n]{\prod_{i=1}^n x_i}$
- GM è l'esponente della media aritmetica dei logaritmi
 - $GM(\mathbf{x}) = e^{AM(\ln \mathbf{x})} = e^{\frac{1}{n} \sum_{i=1}^n \ln x_i}$
 - (si dimostra subito usando le proprietà dei logaritmi)
 - (a volte l'esponente non serve:
 $\frac{1}{n} \sum_{i=1}^n \ln x_i$)

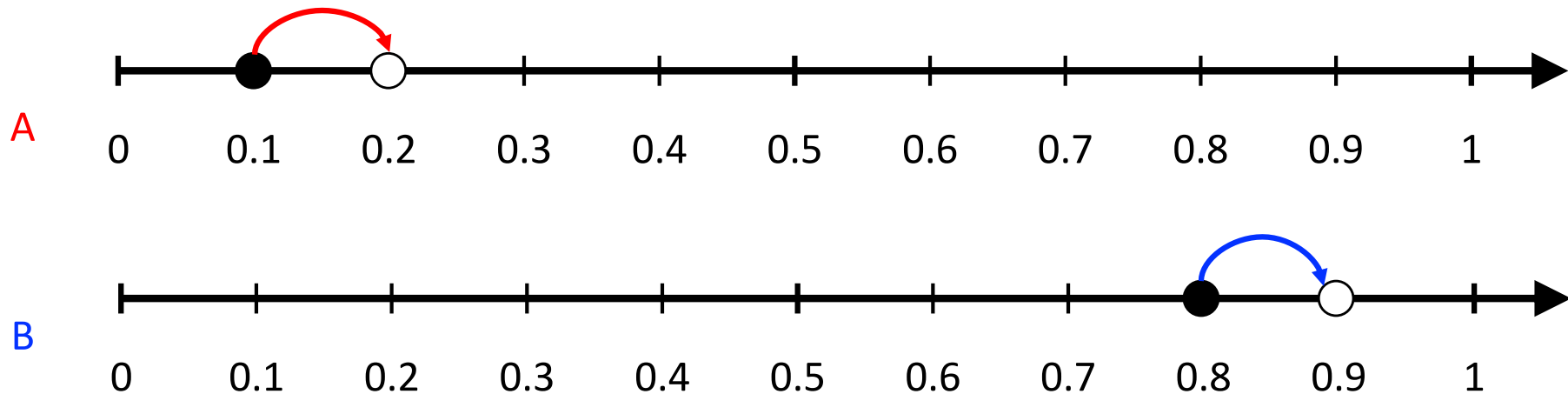
Esempio

- Misurazione di efficacia di qualcosa su scala $[0, 1]$
 - Es.: accuracy di un classificatore, bravura di uno studente, efficacia di un motore di ricerca, ..., ...

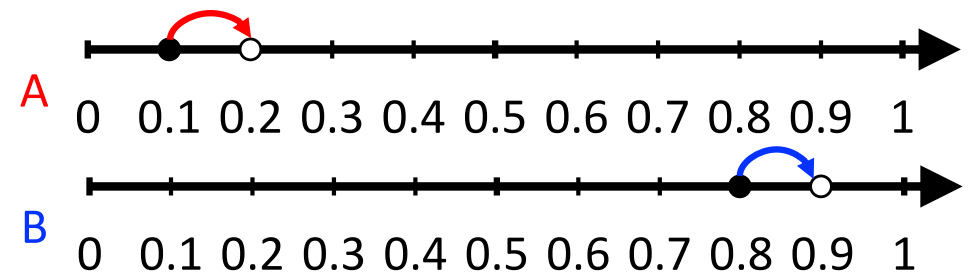


2 casi

- Caso **A** migliore da 0.1 a 0.2
- Caso **B** migliore da 0.8 a 0.9

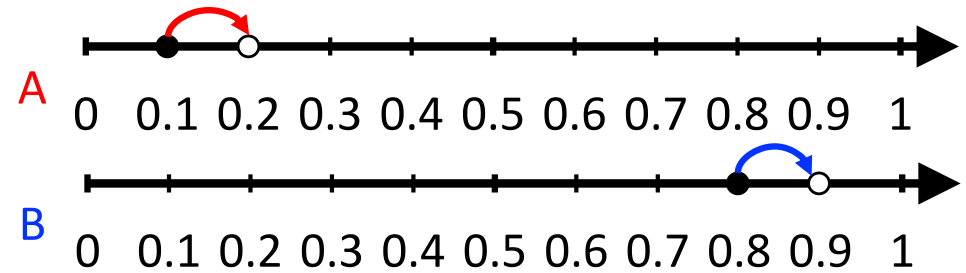


Quale preferite?

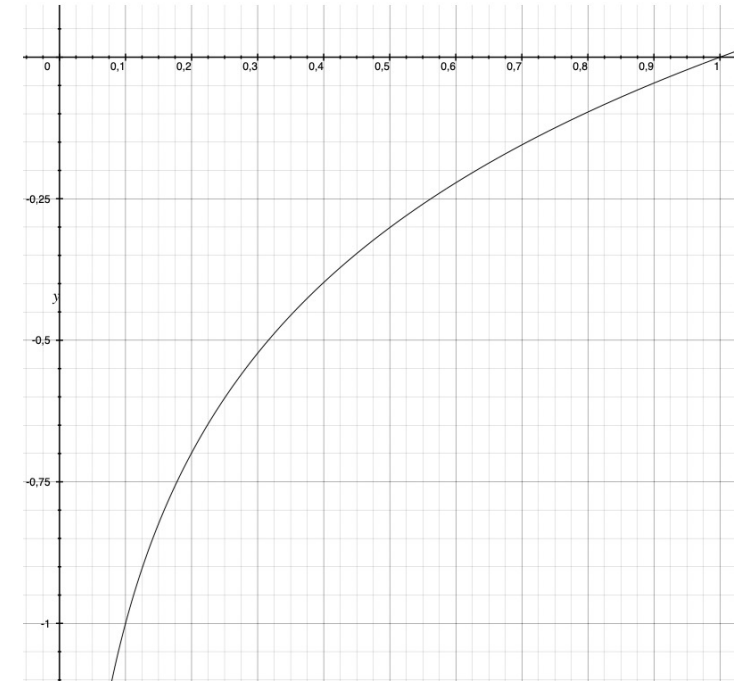


- Ragionamento 1: sono =
 - incremento di 0.1 in entrambi i casi
- Ragionamento 2: meglio il rosso **A**
 - Raddoppia le prestazioni, incremento % >
 - Migliora una cosa che andava molto male
 - E che quindi forse era difficile
- Ragionamento 3: meglio il blu **B**
 - Mi avvicino alla "perfezione"
 - È difficile migliorare qualcosa che è già molto buono

Cosa misurano AM e GM?



- AM li considera =
- GM premia il rosso **A**
 - "stira" lo spazio sui valori più bassi
 - Intervallo rosso = intervallo blu:
 - $|0.2 - 0.1| = |0.9 - 0.8|$
 - Ma
 - $|\log 0.2 - \log 0.1| > |\log 0.9 - \log 0.8|!$
- GM utile per "problemi difficili"
 - Che sono forse i più interessanti



Es. concreto

- Motore di ricerca
- Meglio un MdR che sa "lavorare bene" su query difficili...
- ... o uno che sa lavorare bene su query facili?
- Formulazione alternativa: miglioramento del 50%
 - $0.2 \rightarrow 0.3$ vs. $0.6 \rightarrow 0.9$
 - AM "premia" il 2°
 - GM li considera =

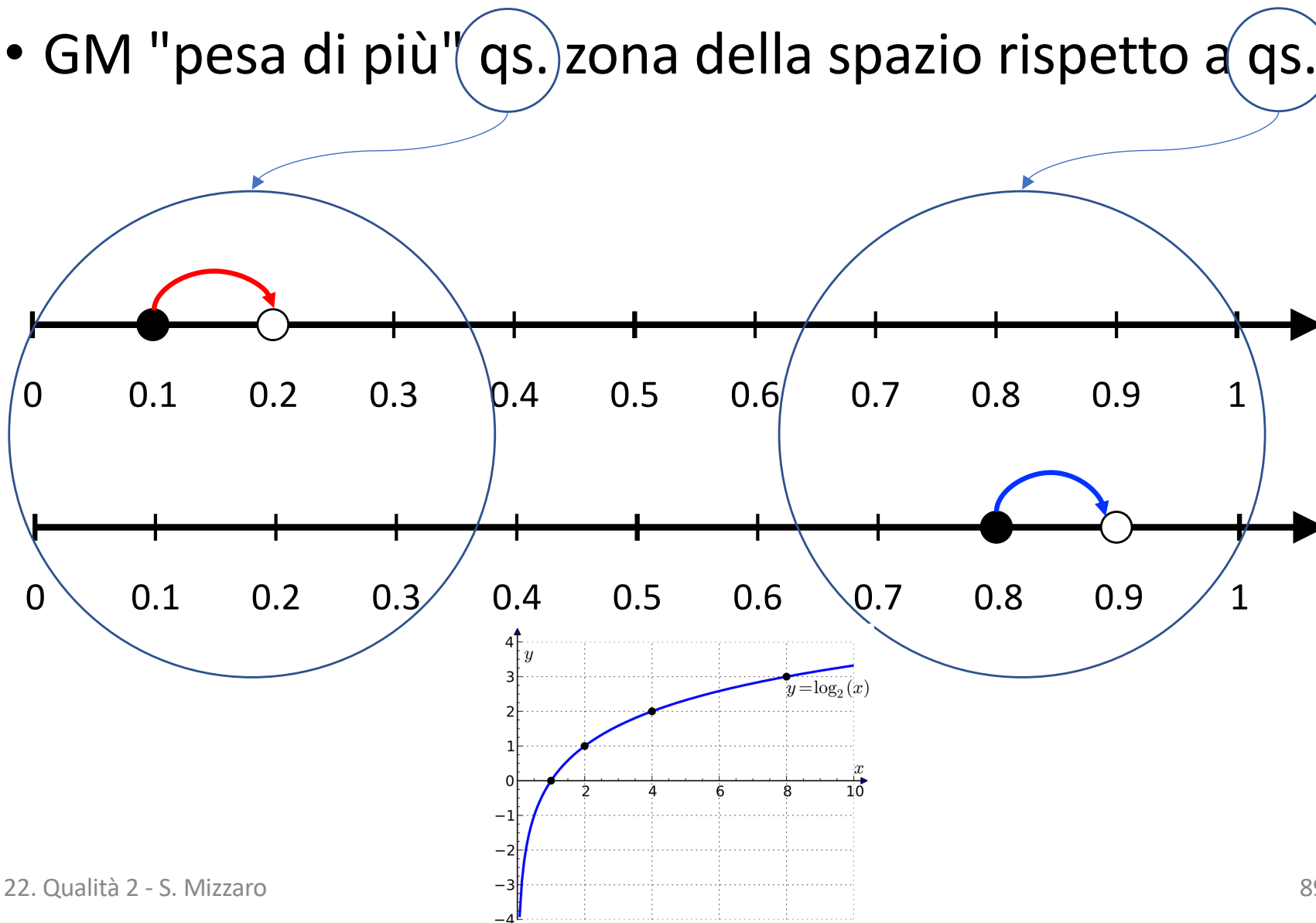
Esempio con più casi

- Item 1 (più "difficile") e item 2, vari sistemi
 - (i1 e i2 potrebbero essere due query a un MdR, ad es.)
- A migliora i2 del 50% e B migliora i1 del 50%
 - AM sale di più con A che con B, GM è neutra
- C vs. A e D vs. A evidenziano come miglioramenti su i1 fanno salire GM di più che su i2

Sistema	Improvement	i1	i2	AM	GM
Baseline		0.2	0.6	0.4	0.35
A	+0.3 (50%) su i2	0.2	0.9	0.55	0.42
B	+0.1 (50%) su i1	0.3	0.6	0.45	0.42
C	+0.2 (100%) su i1	0.4	0.6	0.5	0.49
D	+0.3 (150%) su i1	0.5	0.6	0.55	0.55

"Stirare" / "Deformare" lo spazio

- GM "pesa di più" qs. zona della spazio rispetto a qs.

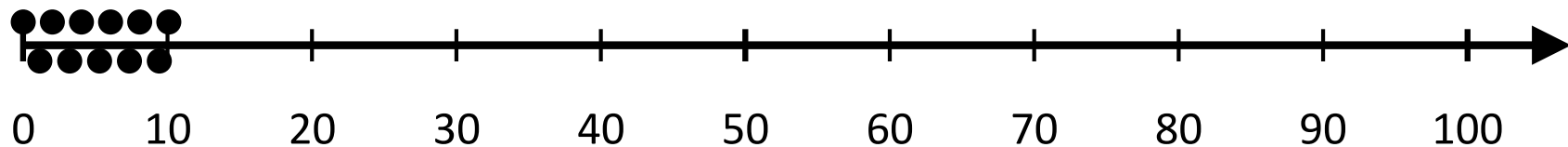


Outlier e AM

- 10+1 worker esprimono un valore in $[0, 100]$

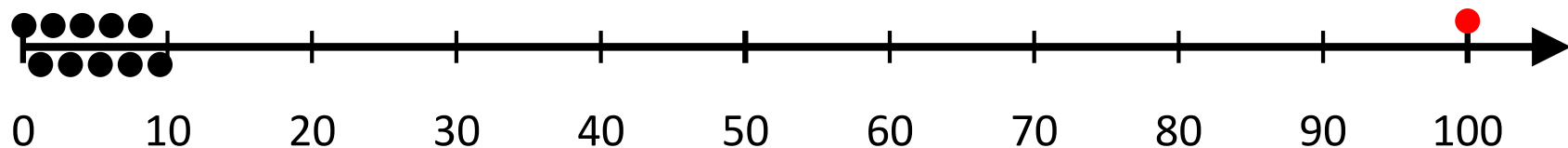
- $(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10)$

- $AM = 5$



- $(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 100)$

- $AM = 13.2$



- E potrebbe essere anche peggio:

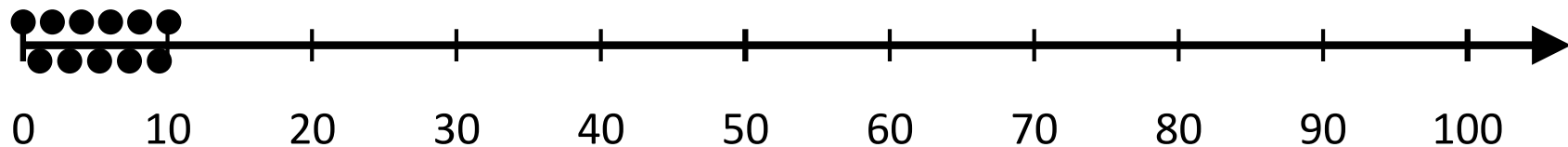
- $(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 1000)$, $AM = 95$

Outlier Con la mediana

- 10+1 worker esprimono un valore in $[0, 100]$

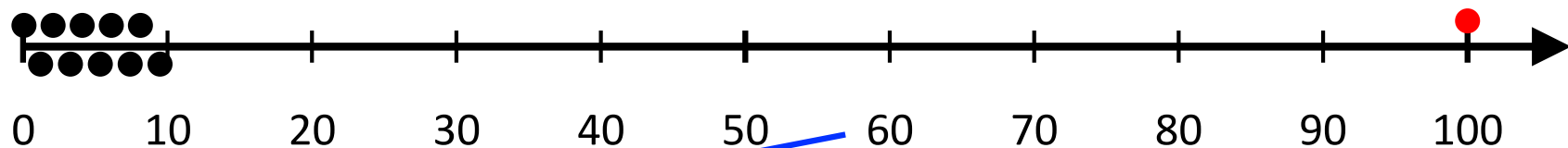
- $(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10)$

- Mediana = 5



- $(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 100)$

- Mediana = 5



- E potrebbe essere anche peggio:

- $(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 1000)$, Mediana = 5

Media armonica (HM)

- "Reciproco della media aritmetica dei reciproci"
- Ha la proprietà di essere "vicina" al min
- Quindi non serve a molto aumentare i valori non min se si vuole aumentare il valore della HM
- (motivo del suo uso in F-measure, F1)
- $$HM(\mathbf{x}) = \frac{1}{\frac{1}{n} \sum_{i=1}^n \frac{1}{x_i}} = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}} = \frac{1}{AM\left(\frac{1}{\mathbf{x}}\right)}$$
- Proprietà:
 - $HM < GM < AM$

Morali

- Ponderare la scelta della funzione di aggregazione a seconda dei casi
 - Pensare al dominio
 - Non fare i nerd 🧐
- Problemi reali
 - Li abbiamo incontrati ad esempio nell'uso di S100
 - Per relevance estimation
 - Per fake news identification
 - GM usata con una tecnica chiamata Magnitude Estimation

"Geothmetic meandian"

$$F(x_1, x_2, \dots, x_n) = \left(\underbrace{\frac{x_1 + x_2 + \dots + x_n}{n}}_{\text{ARITHMETIC MEAN}}, \underbrace{\sqrt[n]{x_1 x_2 \dots x_n}}_{\text{GEOMETRIC MEAN}}, \underbrace{x_{\frac{n+1}{2}}}_{\text{MEDIAN}} \right)$$

$$\text{GMDN}(x_1, x_2, \dots, x_n) = \underbrace{F(F(F(\dots F(x_1, x_2, \dots, x_n) \dots)))}_{\text{GEOTHMETIC MEANDIAN}}$$

$$\text{GMDN}(1, 1, 2, 3, 5) \approx 2.089$$

STATS TIP: IF YOU AREN'T SURE WHETHER TO USE THE MEAN, MEDIAN, OR GEOMETRIC MEAN, JUST CALCULATE ALL THREE, THEN REPEAT UNTIL IT CONVERGES

DIVISION NOTATION

$$\left. \begin{array}{l} A \div B \\ B \overline{)A} \end{array} \right\} \text{SCHOOLCHILD}$$

$$A/B \quad \text{SOFTWARE ENGINEER}$$

$$A/B \quad \text{NORMAL PERSON OR UNICODE ENTHUSIAST}$$

$$\frac{A}{B} \quad \text{SCIENTIST}$$

$$AB^{-1} \quad \text{FANCY SCIENTIST}$$

$$\left. \begin{array}{l} F(A, B) \\ \text{SUCH THAT} \\ F(B) = A \end{array} \right\} \text{OH NO, RUN}$$

Riassunto

- Abbiamo capito alcune cose su
 - Teoria della misurazione
 - Aggregazione
 - (Alcune) Funzioni di aggregazione
 - Media aritmetica e Media geometrica
 - Outlier
 - Mediana
 - Media armonica

Riassunto

- Design del task
 - Ridondanza & Aggregazione
 - Teoria della misurazione
 - Aggregazione reloaded
 - **Accordo**
 - Def., Varie misure
 - Come sfruttarlo per la qualità
 - Un modello più sofisticato
- Lezione scorsa
- Prossima lezione

Riassunto: Tecniche per qualità

- Metodologiche
 - Design del task (Pairwise, scale, ...)
- In itinere
 - Qualification test
 - Gold questions, coerenza
 - Controlli sintattici
 - Monitoraggio del comportamento (tempo, azioni, log, ...)
 - Ridondanza. Aggregare
- Ex-post
 - Pulizia dati (rimozione outlier, ecc.)
 - Aggregazione sofisticata (co-determination algorithms)
 - Sfruttare l'accordo

Biblio

- Misurazione:
 - [Amigó & Mizzaro, MT] "Dispensina" (estratto da articolo)
 - [Stevens 1946] S. S. Stevens, *On the Theory of Scales of Measurement*, Science 103(2684), pp. 677-680
 - Pagina Wikipedia su Measurement
<https://en.wikipedia.org/wiki/Measurement>

SCIENCE

Vol. 103, No. 2684

Friday, June 7, 1946

On the Theory of Scales of Measurement

S. S. Stevens

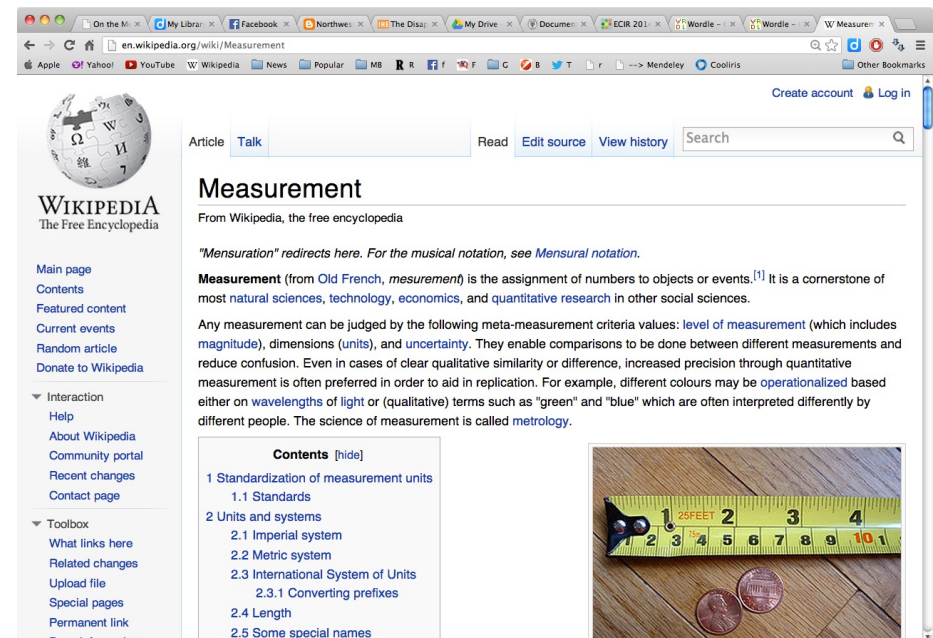
Director, Psycho-Acoustic Laboratory, Harvard University

FOR SEVEN YEARS A COMMITTEE of the British Association for the Advancement of Science debated the problem of measurement. Appointed in 1932 to represent Section A (Mathematical and Physical Sciences) and Section J (Psychology), the committee was instructed to consider and report upon the possibility of "quantitative estimates of sensory events"—meaning simply: Is it possible to measure human sensation? Deliberation led only to disagreement, mainly about what is meant by the term measurement. An interim report in 1938

by the formal (mathematical) properties of the scales. Furthermore—and this is of great concern to several of the sciences—the statistical manipulations that can legitimately be applied to empirical data depend upon the type of scale against which the data are ordered.

A CLASSIFICATION OF SCALES OF MEASUREMENT

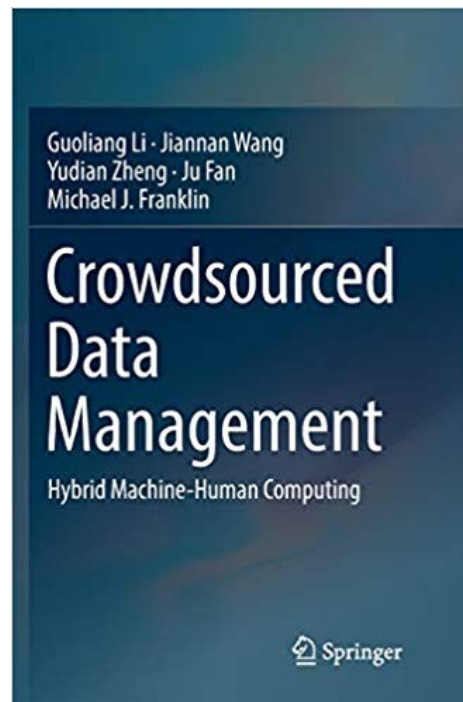
Paraphrasing N. R. Campbell (Final Report, p. 340), we may say that measurement, in the broadest sense, is defined as the assignment of numerals to objects or events according to rules. The fact that



The screenshot shows the Wikipedia page for "Measurement". The page title is "Measurement" and it is part of the "Wikipedia: The Free Encyclopedia". The page content includes a definition of measurement as the assignment of numbers to objects or events, and a list of contents. The contents list includes "1 Standardization of measurement units", "2 Units and systems", "2.1 Imperial system", "2.2 Metric system", "2.3 International System of Units", "2.3.1 Converting prefixes", "2.4 Length", and "2.5 Some special names". There is also a small image of a ruler and coins.

[Li et al, cap. 3]

- Guoliang Li, Jiannan Wang, Yudian Zheng, Ju Fan, Michael J. Franklin. *Crowdsourced Data Management – Hybrid Machine-Human Computing*, Springer, 2018



Biblio

- https://en.wikipedia.org/wiki/Geometric_mean
- https://en.wikipedia.org/wiki/Harmonic_mean

Sull'agreement by chance (e altro)

- https://www.ted.com/talks/hans_and_ola_rosling_how_not_to_be_ignorant_about_the_world/transcript
- <https://www.sli.do/> #T987, #T988, #T989
- Compiti per casa (entro fine vacanze):
 - 1) Guardare il video per intero, ma...
 - 2) ... rispondendo alle 3 domande su slido man mano
 - 3) Poi, riflettere sull'agreement by chance
- Gli highlight secondo me:
 - "So I went to the zoo and I asked the chimps"
 - "Aaahhhh, you almost made it to the chimps"